

Introduction to Spatial Data visualisation and Mapping in R

Statistics Netherlands (CBS)

Martijn Tennekes

Session 4: Exploring Colour Palettes with the R Package cols4all

Introduction to cols4all

Key Features

- **Comprehensive Palette Collection:** Includes a wide range of palettes from popular sources like ColorBrewer, viridis, and scico, as well as custom and user-defined palettes.
- **Palette Classification:** Organizes palettes into types such as categorical, sequential, diverging, cyclic, and bivariate.
- **Dashboard for exploration and analysis:**
 - Colorblind-Friendliness
 - Contrast
 - Fairness
 - HCL Analysis
- **Interactive Dashboard:** Launch with `c4a_gui()`.
- **Integration:** Works with `ggplot2` and `tmap`.



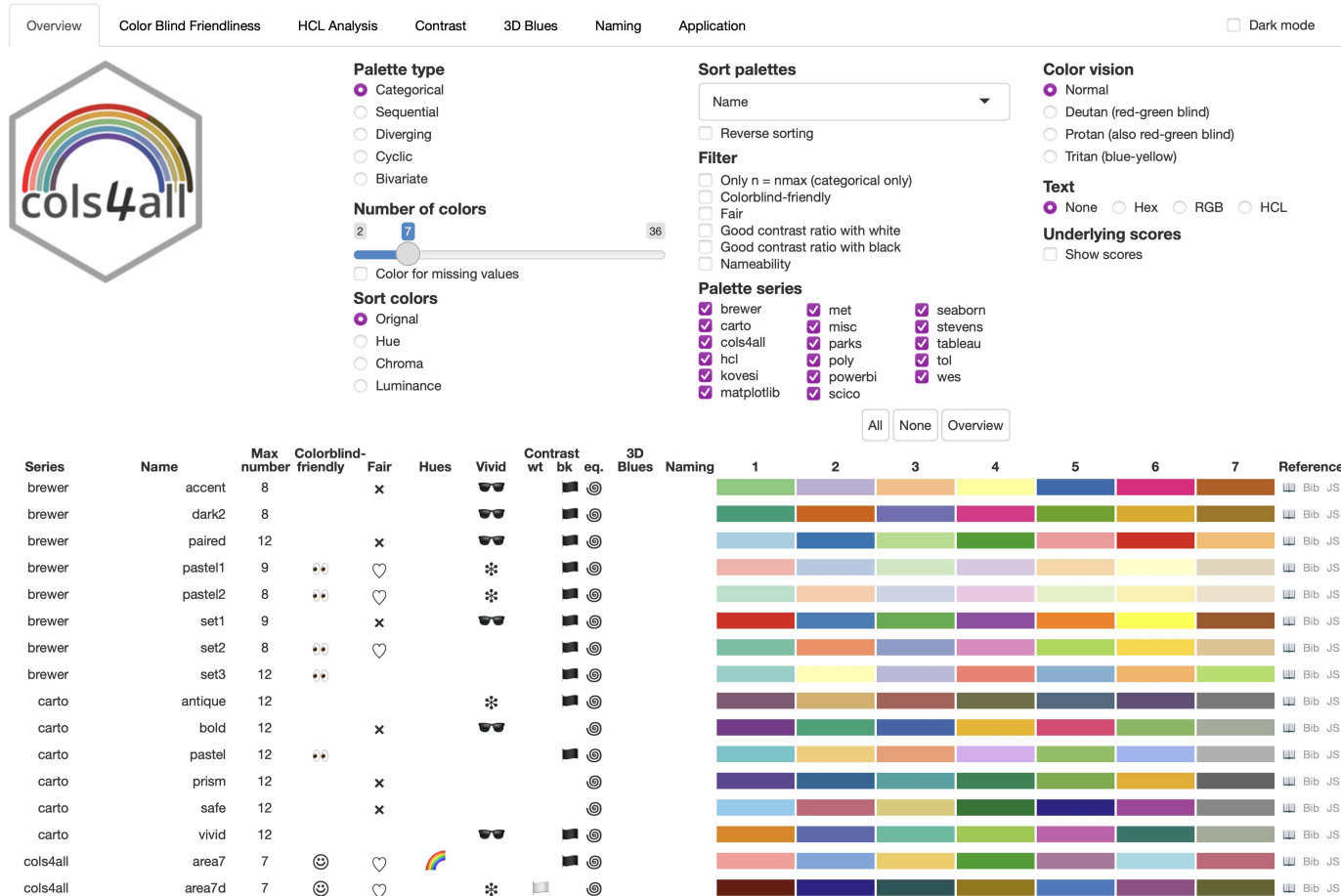
Evaluation Metrics

- **Colorblind-Friendliness:** Uses distance metrics to assess distinguishability under different types of color vision deficiencies.
- **Contrast Ratios:** Ensures readability by calculating contrast between colors and backgrounds.
- **Fairness Metric:** A novel metric assessing visual balance so no color dominates.

Links

- [Documentation](#)
- [GitHub repo](#)

Dashboard



Categorical palette table

Series	Name	Max number	Colorblind-friendly	Fair	Hues	Vivid	Contrast wt	bk	eq.	3D Blues	Naming	1	2	3	4	5	6	7	References
tol	light	9		♥															Bib JS R
parks	crater_lake	7		♥		✱													Bib JS R
seaborn	muted	10		♥															Bib JS R
tableau	classic10light	10		♥															Bib JS R
hcl	dark3	36		♥															Bib JS R
seaborn	deep	10		♥															Bib JS R
hcl	set3	36	👁️	♥		✱													Bib JS R
hcl	dynamic	36	👁️	♥		✱													Bib JS R
tableau	classic_cyclic	13	👁️	♥		🕶️													Bib JS R
brewer	set2	8	👁️	♥															Bib JS R
tableau	hue_circle	19	👁️	♥															Bib JS R
hcl	dark2	36	👁️	♥		✱													Bib JS R
hcl	set2	36	👁️	♥		✱													Bib JS R
brewer	pastel1	9	👁️	♥		✱													Bib JS R
seaborn	pastel	10	👁️	♥															Bib JS R
hcl	harmonic	36	👁️	♥		✱													Bib JS R
brewer	pastel2	8	👁️	♥		✱													Bib JS R
hcl	cold	36	👁️	♥		✱													Bib JS R
hcl	pastel1	36	👁️	♥		✱													Bib JS R
hcl	warm	36	👁️	♥		✱													Bib JS R

Diverging table

Series	Name	Colorblind-friendly	Fair	Vivid	Hues	Contrast wt	bk	3D Blues	1	2	3	4	5	6	7	References
matplotlib	pi_yg	☺			🌓											
hcl	red_green	☺			🌓		🚩									
scico	bam	☺	♥	✳️	🌓		🚩									
scico	vanimo	☺		✳️	🌓											
hcl	purple_green	☺	♥	✳️	🌓		🚩									
brewer	prgn	☺	♥	✳️	🌓											
tol	pu_gn	☺	♥	✳️	🌓		🚩									
matplotlib	prgn	☺	♥	✳️	🌓											
met	cassatt2	☺	♥	✳️												
parks	acadia	☺	✖	🕶️												
hcl	purple_brown	☺	♥	✳️	🌓		🚩									
kovesi	bu_gy_yl	☺			🌓											
matplotlib	seismic	☺	✖	🕶️	🌓			🔵								
kovesi	bu_wh_rd	☺	✖	🕶️	🌓		🚩									
kovesi	bu_wh_rd2	☺	✖	🕶️	🌓		🚩									
matplotlib	coolwarm	☺	✖	🕶️												
kovesi	bu_bk_rd	☺	✖	🕶️	🌓		🚩									
kovesi	bu_bk_br	☺	✖	🕶️	🌓		🚩									

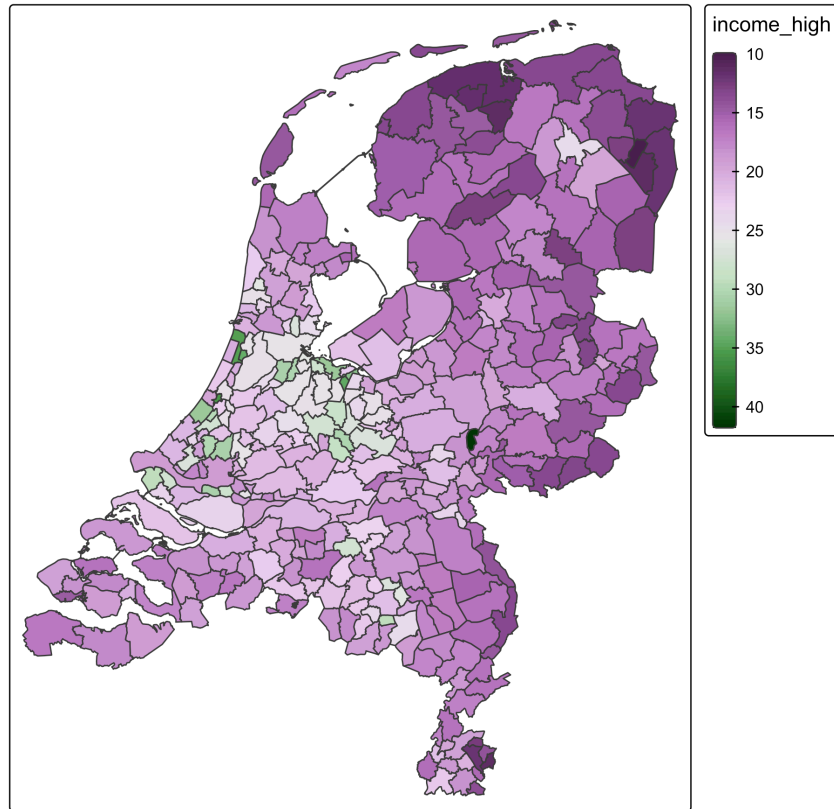
New cols4all palettes

Series	Name	Max number	Colorblind-friendly	Fair	Hues	Vivid	Contrast wt bk eq.	3D Blues	Naming	1	2	3	4	5	6	7	References
cols4all	area7	7	☺	♡	🌈		☼ ☼ ☼										
cols4all	area7d	7	☺	♡		✱	☼ ☼ ☼										
cols4all	area8	8		♡			☼ ☼ ☼										
cols4all	area8d	8	☺	♡		✱	☼ ☼ ☼										
cols4all	area9	9		♡			☼ ☼ ☼										
cols4all	area9d	9		♡			☼ ☼ ☼										
cols4all	friendly11	11	☺			🕶	☼ ☼ ☼										
cols4all	friendly13	13	☺			🕶	☼ ☼ ☼										
cols4all	friendly7	7	☺☺	✕	🌈	🕶	☼ ☼ ☼										
cols4all	friendly9	9	☺☺	✕	🌈	🕶	☼ ☼ ☼										
cols4all	line7	7	☺			🕶	☼ ☼ ☼										
cols4all	line8	8		✕		🕶	☼ ☼ ☼										
cols4all	line9	9				🕶	☼ ☼ ☼										

Integration with tmap and ggplot2

Palettes in tmap

```
tm_shape(NLD_muni) +  
  tm_polygons(fill = "income_high",  
              fill.scale = tm_scale_continuous(values = "cols4all.pu_gn_div"))
```



Palettes in ggplot2

```
data("diamonds")
diam_exp = diamonds[diamonds$price >= 15000, ]
diam_exp$clarity[1:500] = NA

# continuous diverging scale
ggplot(diam_exp, aes(x = carat, y = depth, color = price)) +
  geom_point(size = 2) +
  scale_color_continuous_c4a_div("cols4all.pu_gn_div", mid = mean(diam_exp$price)) +
  theme_light()
```



Recap

- **cols4all** is a handy R package for color palette exploration
- [Main source of documentation](#)